

Ismam Nur Swapnil

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Education

Bangladesh University of Engineering and Technology
BSc in Electrical and Electronic Engineering

Dhaka, Bangladesh
Feb 2020 – Feb 2025

- **CGPA:** 3.73/4.0
- **Relevant Coursework:** Linear Algebra, Probability and Statistics, Artificial Intelligence and Machine Learning, Robotics and Automation, Digital Image Processing, Random Signals and Processes, Control Systems, Digital Signal Processing, Continuous Signals and Linear Systems.

Research Interests

- Reinforcement Learning
- Probabilistic Learning
- Multimodal Large Language Models
- Optimization Methods and Algorithms
- Physics Informed Learning
- Data-Efficient Learning

Publications

- **GRPO++: Enhancing Dermatological Reasoning under Low Resource Settings**
I. N. Swapnil, A. Saha, T. A. Khan, & M. A. Haque
Under review at IEEE Journal of Biomedical and Health Informatics — [Preprint](#)
- **CLARIFY: A Specialist-Generalist Framework for Accurate and Lightweight Dermatological Visual Question Answering**
A. Saha, T. A. Khan, I. N. Swapnil, & M. A. Haque — [Preprint](#)
- **Compression Strategies for Efficient Multimodal LLMs in Medical Contexts**
T. A. Khan, A. Saha, I. N. Swapnil, & M. A. Haque — [Preprint](#)

Research Experience

University of Central Florida (UCF)
Research Collaborator

Orlando, FL (Remote)
Nov 2025 – Present

- Collaborating with **Prof. Ser-Nam Lim** on the optimization of Policy Gradient algorithms for Large Language Models.
- Researching **GRPO variants** for enhanced convergence stability and **mode collapse mitigation** during model reasoning.

Bangladesh University of Engineering and Technology
Undergraduate Researcher

Dhaka, Bangladesh
Feb 2024 – Feb 2025

- Developed **GRPO++**, a stabilized Group Relative Policy Optimization method that enhances reasoning capabilities in Vision-Language Models under low-resource settings, followed by DPO alignment using a Knowledge Graph-based preference system.
- Designed the **CLARIFY framework**, a specialist-generalist architecture for lightweight and efficient medical visual question answering.

Independent Research

Feb 2025 – Present

- Researching **inference-time scaling** and **advanced sampling strategies** to enhance reasoning capabilities in Large Language Models, with a focus on improving the efficiency, sample complexity, and data utilization of **online Reinforcement Learning algorithms**.

Professional Experience

Advanced Chemical Industries Limited
Machine Learning Engineer, Department of MIS

Dhaka, Bangladesh
Jun 2025 – Present

- Developed **AgroResearch AI**, an intelligent chatbot for agricultural research retrieval and information synthesis.
- Currently working on the development of a **Bangla Text-to-Speech (TTS)** system for natural and expressive Bangla audio synthesis.
- Working on the **Bangla OCR** project to enable accurate optical character recognition for any kind of Bangla Documents.

Technical Skills

- **Generative AI:** Multimodal LLM, RAG, Agents, Fine-Tuning (SFT, QLoRA, DPO), RLHF, Model-pruning
- **ML-DL:** Deep Learning, Computer Vision (Feature Extraction, Object Detection, Segmentation)
- **Programming Languages:** Python, MATLAB
- **ML Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn
- **Agent Development:** LangChain, LangGraph, LlamaIndex
- **API Development:** FastAPI
- **DevOps Tools:** Docker, Git
- **Data Processing:** Pandas, NumPy, OpenCV, SciPy

Notable Projects

- **AgroResearch AI** 
Built an intelligent chatbot for agricultural research and information retrieval. The system uses RAG-based retrieval to synthesize insights from scientific literature and provide actionable recommendations for farming practices and crop/soil/cattle/aqua etc management.
- **Bangla OCR Pipeline** 
Developed an end-to-end OCR pipeline for Bangla handwritten and printed text, combining YOLO for word detection and Qwen3-VL-2B for accurate text recognition. Supports both PyTorch and ONNX backends for flexible deployment.
- **Real-Time AI Sales Communicator** 
Developed a real-time AI Voice Sales Agent using Python, LangChain, FastAPI, and WebSockets. Engineered a full-duplex capable system integrating local Qwen LLM for conversational AI, Whisper for STT, and pyttsx3 for TTS.
- **Skin Disease Classification using Concatenated Deep Learning Models** 
Developed an advanced deep learning model by combining multiple pre-trained networks to classify various skin diseases such as eczema, psoriasis, and melanoma. Improved classification accuracy using ensemble learning techniques.

Honors and Awards

- **University Merit Scholarship**, Bangladesh University of Engineering and Technology (BUET) — Awarded for academic excellence in 2 semesters.
- **Dean's List Award**, Bangladesh University of Engineering and Technology (BUET) — Recognized for achieving a GPA of 3.75+.
- **139th Position in BUET Undergraduate Admission Test** — Secured position among over 12,000 candidates.

Certifications & Achievements

- **Undergraduate Thesis Poster** 
Presented thesis on AI-driven dermatological diagnosis using Vision-Language Models, fine-tuned with SFT, QLoRA, and DPO techniques.
- **Runner Up – Poster Presentation** 
Awarded Runner Up at Bear Summit 2024 for presenting research on Vision-Language Models for dermatological diagnosis. Organized by Department of EEE, BUET.
- **Participant – National Semiconductor Symposium 2024** 
Attended the national symposium featuring industry leaders and academic researchers discussing advancements in semiconductor technology and AI hardware.

References

Dr. Mohammad Ariful Haque

Professor, Department of EEE

Bangladesh University of Engineering and Technology

Email: arifulhoque@eee.buet.ac.bd

Relationship: Undergraduate Thesis Supervisor

Dr. Mohammad Jahangir Alam

Professor, Department of EEE

Bangladesh University of Engineering and Technology

Email: mjalam@eee.buet.ac.bd

Relationship: Undergraduate Academic Advisor